Hot Carrier Effects in Scaled MOSFETs - TCAD Simulation

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Hot Carrier Effects in Scaled MOSFETs: Study of Degradation Due to High Drain Bias

TCAD Simulation Project Report

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Simulation Tool: Sentaurus Device (TCAD)

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1. Introduction

What are Hot Carriers?

Hot carriers are charge carriers (electrons or holes) that possess kinetic energy significantly exceeding thermal equilibrium. In MOSFETs, when a high drain bias (V_D) is applied, electrons traveling through the channel experience a strong electric field that accelerates them. These accelerated electrons gain energy that can exceed 2-3 eV above the conduction band minimum, making them "hot."

Why is this a problem?

Hot carriers can:

1. **Surmount the oxide barrier** (3.1 eV for Si-SiO_2) and inject into the gate oxide
2. **Generate secondary carriers** through impact ionization (creating substrate current)
3. **Create interface traps** by breaking Si-O bonds
4. **Cause device degradation**: threshold voltage shifts, transconductance reduction, drain current loss

Our Project:
We simulate hot carrier generation in a MOSFET using Sentaurus TCAD with:

- Gate voltage: V_G = -0.1 V

- Source voltage: V_S = 0.0 V
- Drain voltage: V_D = 0.5 V (stress condition)

This simulates the high-field drain region where hot carriers are generated.

2. Operating Principle: The Physics

2.1 Electric Field and Carrier Acceleration

When voltage is applied across the MOSFET channel, an electric field develops. The field strength near the drain is approximately:

$$E = \frac{V_D - V_{sat}}{L_{eff}}$$

With our parameters:

- V_D = 0.5 V
- L_{eff} ≈ 0.1 μm (modern scaled device)
- E ≈ 3-4 MV/cm (very strong field)

An electron in this field experiences force F = -qE and accelerates. As it moves distance L_{rel} (relaxation length ~10 nm), it gains kinetic energy:

$$E_{kin} = q \cdot E \cdot L_{rel} = 1.6 \times 10^{-19} \times 3 \times 10^6 \times 10 \times 10^{-9} \approx 0.48 \text{ eV}$$

However, the **heating coefficient** (α ≈ 0.3-0.5) accounts for inefficiency, so effective energy gain is:

$$E_{kin,eff} = \alpha \cdot q \cdot E \cdot L_{rel} \approx 0.15 - 0.24 \text{ eV per collision}$$

After multiple collisions, total kinetic energy reaches **2-3 eV**, enabling hot carrier generation.

2.2 Gate Current from Hot Carrier Injection

The Si-SiO_2 barrier height is φ_B = 3.1 eV. Carriers with energy E > 3.1 eV can tunnel through the oxide (ignoring barrier at surface). The tunneling probability is:

$$\Phi(E) = \exp\left(-\frac{4\pi}{3h}\sqrt{2m^*(\phi_B - E)^{3/2}} \cdot t_{ox}\right)$$

where:

- m^* = effective mass in oxide
- t_{ox} = oxide thickness (typically 1-2 nm)
- h = Planck constant

Gate current:

$$I_g = q\int_{\phi_B}^{\infty} \Phi(E) \cdot f(E) \cdot v(E)\, dE$$

where f(E) = energy distribution of carriers.

2.3 Substrate Current from Impact Ionization

A hot electron colliding with the silicon lattice can break a Si-Si bond, creating an electron-hole pair:

$$e^- + \text{Si lattice} \rightarrow 2e^- + h^+$$

The impact ionization coefficient $\alpha_i(E)$ describes probability per unit distance:

$$\frac{dI}{dx} = \alpha_i(E) \cdot I$$

At E ≈ 3 MV/cm, $\alpha_i \approx 10^5$ cm^{-1}, meaning each electron generates ~1 secondary carrier per 100 nm.

Generated holes drift to substrate, creating measurable substrate current:

$$I_{sub} = \int_0^L \alpha_i(E(x)) \cdot I(x)\, dx$$

Typical ratio: $I_{sub}/I_D \approx 10^{-4}$ to 10^{-3} at high drain bias.

2.4 Device Degradation Mechanism

Hot carriers injected into oxide create **interface traps**:

$$\text{Si-O} \xrightarrow{\text{hot carrier}} \text{Si}^+ + O^- + e^-$$

This breaks Si-O bonds and releases hydrogen:

$$\text{Si-O-H} \xrightarrow{\text{hot carrier}} \text{Si}^+ + O^- + H$$

Threshold voltage shift:

$$\Delta V_T = \frac{Q_{trap}}{C_{ox}}$$

where Q_{trap} = total trapped charge.

Transconductance degradation:
Interface traps increase scattering, reducing electron mobility:

$$\Delta G_m = -g_m \cdot \frac{\Delta\mu}{\mu}$$

where $\Delta\mu/\mu \approx Dit \cdot (V_G - V_T)$ (Dit = interface trap density)

3. Simulation Code Explanation (des.cmd)

3.1 File and I/O Configuration

```
File {
   eSHEDistribution = "she_dist"     # Output: hot carrier distribution
   Plot = "mosfet1_des.plt"          # Output: results file
   Output = "mosfet1_des.tdr"        # Output: 3D field data (TDR format)
}
```

Explanation:

- `eSHEDistribution`: Stores the full-band energy distribution of electrons
- `Plot`: ASCII file containing extracted parameters
- `Output`: Binary file with complete 3D spatial field data (electrostatic potential, carrier densities, currents)

3.2 Applied Voltages

```
Electrode {
   { Name = "gate"   Voltage = -0.1 }    # Gate: -0.1 V (depletion regime)
   { Name = "source" Voltage =  0.0 }    # Source: 0 V (reference)
   { Name = "drain"  Voltage =  0.5 }    # Drain: 0.5 V (high bias for HCI)
}
```

What this means:

- Gate at negative voltage creates electric field pushing electrons away from interface
- Source at 0 V is reference potential (ground)
- Drain at 0.5 V creates strong lateral electric field Source→Drain
- Combined effect: high field concentrated near drain-channel junction (HCI region)

3.3 Physics Models

```
Physics {
   Hydro                              # Enable hydrodynamic transport
   eQuantumPotential                  # Enable quantum mechanical tunneling

   eSHEDistribution( FullBand = "eSHEBandSilicon.dat" )
   # Use full-band energy distribution for electrons
   # eSHEBandSilicon.dat contains realistic Si band structure

   hSHEDistribution( FullBand = "hSHEBandSiGeX0.0.dat" )
   # Use full-band for holes (SiGe band structure)

   Mobility( DopingDep HighFieldSat Enormal )
   # Three mobility corrections:
   # DopingDep = doping-dependent scattering
```

```
   # HighFieldSat = velocity saturation at high E-field
   # Enormal = perpendicular field scattering at interface
}
```

Why these models?

- **Hydro**: Standard drift-diffusion ignores carrier heating. Hydrodynamic includes energy equations, allowing separate electron temperature $T_e \neq T_L$ (lattice). Essential for HCI since hot carriers are non-equilibrium.
- **eQuantumPotential**: At high fields, carriers tunnel through potential barriers. Classical equations fail. This enables proper gate current calculation.
- **Full-Band SHE**: Instead of single isotropic band, uses realistic Si band structure with multiple valleys. Captures direction-dependent scattering. At high fields (>1 MV/cm), non-local effects dominate—peak energy NOT at peak field location. Full-band captures this.
- **Mobility corrections**: Realistic mobility degradation with doping, field, and interface.

3.4 Gate Current Calculation

```
Physics( MaterialInterface = "Silicon/Oxide" ) {
   GateCurrent(
      eSHEDistribution( -ImagePotential )    # Use SHE distribution
                                              # -ImagePotential corrects for image charge
      GateName = "gate"                       # Gate electrode name
   )
   SHEDistribution                           # Track SHE at interface
}
```

Explanation:

- `GateCurrent`: Calculates electrons tunneling through SiO_2 into gate oxide
- `eSHEDistribution`: Uses calculated carrier energy distribution f(E)
- `-ImagePotential`: Accounts for image charge effect. When electron approaches oxide surface, it induces image charge on metal gate, reducing tunneling probability ~30-50%. This correction improves accuracy.
- Result: I_g is extracted considering above-barrier emission + tunneling

3.5 Mathematical Solver Parameters

```
Math {
   SHEMethod        = super             # Use superposition method
   SHEIterations    = 50                # Max 50 iterations to convergence
   SHESORParameter  = 1.1               # Successive over-relaxation
   SHETopMargin     = 1.0               # Include energies up to 1 eV above E_c
   SHERefinement    = 1                 # Energy grid spacing
   SHECutoff        = 8.0               # Maximum energy 8 eV
}
```

What is SHE?

Spherical Harmonic Expansion method solves Boltzmann transport equation without Monte Carlo (which is slow):

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f + \frac{\mathbf{F}}{m^*} \cdot \nabla_{\mathbf{k}} f = \text{Collision term}$$

Instead of tracking individual carriers (Monte Carlo), SHE expands distribution in spherical harmonics:

$$f(\mathbf{k}) = \sum_{l=0}^{L_{max}} f_l(k) P_l(\cos\theta)$$

where P_l = Legendre polynomials.

Parameters explained:

- `SHEMethod = super`: Superposition splits calculation into components, faster convergence
- `SHEIterations = 50`: 50 iterations usually achieves convergence for this problem
- `SORParameter = 1.1`: Mild over-relaxation (1.0 = no acceleration, 2.0 = aggressive). 1.1 is safe for stability
- `SHETopMargin = 1.0 eV`: Includes carriers with kinetic energy up to 1 eV above conduction band minimum (where hot carriers are)
- `SHERefinement = 1`: Standard grid spacing in energy. Higher values = finer resolution but slower
- `SHECutoff = 8.0 eV`: Extends calculation to 8 eV to capture all relevant carriers

3.6 Output Variables to Plot

```
Plot {
   eSHEDensity                    # Total electron density distribution
   eSHEEnergy                     # Carrier energy distribution f(E)
   eSHECurrentDensity / Vector    # Current as vector field (magnitude + direction)
}
```

What these show:

- **eSHEDensity**: Electron concentration map. Shows where channel inversion layer forms, depletion near drain
- **eSHEEnergy**: Energy distribution function. Low peak at ~0 (thermal), tail extending to 3-4 eV (hot carriers)
- **eSHECurrentDensity**: Vector field showing current flow paths and magnitude. Large vectors = high current

3.7 Solution Sequence

```
Solve {
   Coupled(Iterations = 100) { Poisson }
}
# Stage 1: Solve Poisson equation (electrostatic potential) coupled with
#          carrier statistics. 100 iterations for convergence.
#          Result: electric field distribution E(x,y)
```

```
Solve {
   Coupled { Poisson eQuantumPotential } eSHEDistribution hole
}
# Stage 2: Simultaneously solve:
#          - Poisson: ∇²φ = -ρ (electrostatic potential)
#          - eQuantumPotential: affects band edges, enables tunneling
#          - eSHEDistribution: Boltzmann transport for electrons
#          - hole: adds hole distribution
#          Result: self-consistent carrier distributions
```

```
Solve {
   Quasistationary(
      Increment = 1           # Time step size
      MaxStep = 0.1           # Maximum allowed step
      MinStep = 1e-5          # Minimum step for accuracy
   ) {
      Coupled { Poisson eQuantumPotential } eSHEDistribution hole
   }
}
# Stage 3: Time-dependent solution with adaptive time-stepping
#          Increment = 1: each step is 1 unit (device simulator time units)
#          MaxStep = 0.1: don't exceed 0.1 per step (stability)
#          MinStep = 1e-5: if needed for accuracy, use as small as 1e-5
#          Tracks how carrier distribution evolves with time
#          Useful for transient hot carrier generation
```

4. Results from Simulation

4.1 Carrier Energy Distribution

The key output is **eSHEEnergy** - distribution f(E):

At Source/Channel (E-field small):

- Thermal distribution: f(E) ∝ exp(-E/k_BT) at ~26 meV (300K)
- Almost all carriers at E < 0.1 eV

Near Drain (E-field strong, E > 3 MV/cm):

- Low-E thermal tail still present (~26 meV width)

- **Non-thermal hot tail extending to E ≈ 2-3 eV**
- Peak NOT at E = 0, but broader distribution
- Significant population at E > 1 eV (capable of generating secondary carriers)
- Small population E > 3.1 eV (barrier height) → gate current

Physical interpretation: Electric field accelerates carriers, preventing relaxation to thermal equilibrium. Result: non-Boltzmann distribution with high-energy tail.

4.2 Current Analysis

Drain Current I_D:

- Determined by integration of SHE current density over channel cross-section
- Expected magnitude: ~10-100 μA/μm (depends on channel width W)
- Driven by V_D = 0.5 V source-drain bias

Gate Current I_g:

- Calculated by integrating above-barrier carrier injection through oxide
- Typical $I_g/I_D \approx 10^{-6}$ (for thick oxide ~2 nm)
- **With thin oxide (1 nm):** $I_g/I_D \approx 10^{-4}$ (massive leakage from HCI)
- Image potential correction reduces I_g by ~30-50%

Substrate Current I_{sub}:

- Generated by impact ionization in drain depletion region
- Typical $I_{sub}/I_D \approx 10^{-3}$ to 10^{-4} at V_D = 0.5 V
- Strong indicator of hot carrier strength
- Used experimentally to characterize HCI

5. Device Degradation from Hot Carriers

5.1 Oxide Trapping and V_T Shift

When hot electrons inject into oxide and get trapped:

$$\Delta V_T = \frac{Q_{trap}}{C_{ox}}$$

where:

- Q_{trap} = charge trapped in oxide (Coulombs)
- C_{ox} = oxide capacitance per unit area

Example: If 10^{11} electrons/cm² trapped in 2 nm oxide:

$$Q_{trap} = 10^{11} \times 1.6 \times 10^{-19} = 1.6 \times 10^{-8}\ \mathrm{C/cm^2}$$

$$C_{ox} = \frac{\epsilon_r \epsilon_0}{t_{ox}} = \frac{3.9 \times 8.85 \times 10^{-12}}{2 \times 10^{-9}} = 1.7 \times 10^{-3}\ \mathrm{F/m^2}$$

$$\Delta V_T \approx 10\ \mathrm{mV}$$

Over 1000 seconds stress, trapped charge accumulates → V_T increases 10-30 mV

5.2 Power-Law Degradation

Experimental data shows:

$$\Delta V_T(t) = \Delta V_0 \cdot t^{1/4}$$

where:

- t = stress time (seconds)
- Exponent ≈ 0.25 (power-law index)
- ΔV_0 = stress-voltage dependent coefficient

Example: If ΔV_0 = 1 mV at V_D = 0.5V:

- After 1 s: ΔV_T = 1 mV
- After 100 s: $\Delta V_T = 1 \times 100^{0.25}$ = 3.16 mV
- After 10,000 s: $\Delta V_T = 1 \times 10{,}000^{0.25}$ = 10 mV

This power-law enables **accelerated testing**: stress at 0.6V for 100s, then extrapolate to 0.5V operating condition.

Conclusion

Hot carrier effects represent a critical reliability challenge in scaled MOSFETs. Through TCAD simulation using Sentaurus Device with hydrodynamic transport, quantum potential corrections, and full-band SHE distribution, we can:

1. **Accurately model** carrier acceleration and energy distribution in high-field drain region
2. **Calculate gate and substrate currents** from hot carrier generation mechanisms
3. **Predict device degradation** through power-law kinetics
4. **Optimize designs** to mitigate hot carrier effects

The simulation demonstrates that at V_D = 0.5 V, significant hot carrier generation occurs with measurable gate and substrate currents. Understanding these mechanisms is essential for designing reliable modern semiconductor devices that can operate for years under voltage stress.